

Alexei Ledenev

Principal Software Architect — Agentic SDLC, Developer Platforms & Multi-Cloud SaaS

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Languages: Hebrew, English

SUMMARY

Principal architect and hands-on engineer with **25+ years** building enterprise multi-tenant SaaS, developer platforms, data platforms, and the tooling engineers use daily.

Current focus: **agentic AI across cloud operations and the software development lifecycle** — multi-agent systems, MCP tooling, safety guardrails, and human-approved execution across AWS, GCP, and Azure. Creator and maintainer of widely adopted open-source tools including **Pumba** and MCP servers for Kubernetes and AWS; former AWS Solutions Architect and HPE Chief Architect at **Distinguished Technologist level**; patent holder, speaker, and hands-on technical leader.

PROFESSIONAL EXPERIENCE

Principal Software Engineer — DoiT International | April 2025 – Present

Leading Reflex, DoiT's strategic agentic AI initiative for cloud operations, while steering multi-tenant SaaS platform migration and analytics modernization.

- Architecting migration of the **DoIT Cloud Intelligence (DCI)** platform from GCP to AWS/EKS for a multi-tenant enterprise FinOps SaaS product.
- Redesigning the analytics pipeline from **BigQuery to ClickHouse**; building the Go sync service for a 6-shard × 2-replica ClickHouse cluster.
- Driving BigQuery cost and performance optimization across the platform, reducing annual spend by **\$100K+**.

Staff Cloud Architect — DoiT International | September 2019 – April 2025

Led a distributed team of senior Forward Deployed Engineers and owned architecture governance across AWS, GCP, Azure, AI/ML, containers, analytics, infrastructure, and automation.

- Led architecture reviews, technical strategy, and best-practice governance across distributed engineering, field, and customer teams — driving technical direction without direct reporting authority.
- Designed and shipped AI products integrated with DoiT Cloud Intelligence: **Zenrouter** for AI-driven ticket routing and an **LLM + OCR invoice-processing workflow** for AWS invoices.
- Drove cloud-native and FinOps adoption across customer engagements; delivered certified AWS/GCP training and spoke at cloud, Kubernetes, DevOps, and agentic-AI events.

Solutions Architect — Amazon Web Services | July 2018 – September 2019

- Ran workshops, immersion days, and reference-architecture sessions.
- Specialized in Kubernetes and container platforms as a member of the **Containers Technical Field Community**.

Chief of Research — Codefresh | March 2017 – July 2018

Codefresh builds a container-native CI/CD and developer-workflow SaaS platform.

- Drove the platform shift from Docker-based CI/CD to a Kubernetes-based build and management platform.
- Improved build performance and developer experience: on-premise/hybrid build runners and Docker layer caching shared across builds.
- Maintained open-source projects and grew the Containers 101 TLV meetup community.

Chief Software Architect — Hewlett-Packard Enterprise, ADM Business Unit | 2013 – March 2017

- Led and mentored product architects across multiple enterprise products.
- Led innovation research and was awarded **9 patents** in test automation, including Mirror Testing and AI-driven Visual Object Recognition for UI automation.
- Drove migration of legacy enterprise products to SaaS, led acquisition due diligence, and supported the organization-wide Agile transition that reduced time-to-market by ~30%.

Earlier Roles | Software Engineer → Team Lead → R&D Manager → Architect | 1997–2013

Progression across Hewlett-Packard, Mercury Interactive, Mer Technologies, and Myriad Interactive from hands-on engineering into technical leadership, R&D management, and architecture.

Selected foundational engineering — Mercury Interactive, TestDirector / Quality Center

- Designed core infrastructure for **Open Test Architecture (OTA)**, the programmable TestDirector SDK that later became Quality Center and HP ALM; established the UI/SDK boundary for test automation, defect tracking, reporting, workflow, and integrations.
- Built cross-language RPC over HTTP between ActiveX/native C++ clients and a Java application server, preserving the SDK's object-oriented model with generated **C++/ATL proxies** and **Java bindings/dispatch code**.
- Invented **FREC**, a recursive, length-prefixed binary serialization format for nested, extensible data structures.

OPEN SOURCE & COMMUNITY

- **Pumba** (Go, **3K+★**) — Creator and maintainer. Chaos engineering for containers: network emulation, fault injection, and stress/performance testing across Docker, Podman, and containerd; used in production resilience testing by engineering teams worldwide.
- **MCP servers for AI-driven cloud operations** — Creator of [k8s-mcp-server](#) and [cloud-mcp-server](#) (Python), letting AI assistants like Claude and Cursor securely run `kubectl`, `helm`, and AWS CLI workflows from hardened, sandboxed environments with scoped credentials.
- **Agentic development tooling** — [ccgram](#), a Telegram ↔ tmux control plane for AI coding agents (Claude Code, Codex CLI, Gemini CLI); [cc-thingz](#), a cross-platform plugin suite packaging skills, agents, and hooks for agentic software development.
- **Cloud-native OSS**: [KubelP](#) for static Kubernetes node IPs; [Secrets-Init](#) for AWS/GCP secret injection; [Spotinfo](#) for AWS Spot inventory and pricing.
- **Technical writing and speaking** — Medium/ITNEXT articles and conference talks on AI coding agents, Kubernetes operations, cloud security, chaos engineering, and cloud-native architecture.

SKILLS

- **AI / Agentic Systems**: LLM application architecture, Model Context Protocol (MCP), multi-agent systems, safety guardrails, evaluation, LangChain, AutoGen, prompt and instruction design
- **Cloud / Platform**: AWS, GCP, Azure, Kubernetes, EKS, GKE, Docker, Cloud Run, BigQuery, ClickHouse, FinOps analytics
- **Engineering**: Go, Python, TypeScript/Node.js, Java, C++, SaaS platforms, distributed systems, RPC, binary serialization, code generation, SDK/API design, microservices, event-driven architectures (GCP Pub/Sub, AWS SQS), platform engineering, internal developer tooling, modular architecture
- **DevOps / Leadership**: Terraform, GitHub Actions, CI/CD, least-privilege IAM, secure sandboxed execution, tenant-aware SaaS design, distributed teams, hiring, mentoring, architecture governance, customer advisory

EDUCATION, CERTIFICATIONS & PATENTS

- **Education**: M.Sc. Computer Science, Tel Aviv University; B.Sc. Computer Science & Statistics, Tel Aviv University
- **Certifications**: GCP Professional Cloud Architect & Authorized Trainer · AWS Solutions Architect – Professional & Authorized Instructor · CKA
- **Patents**: 9 granted patents in test automation, UI testing, protocol handling, and AI-driven automation.